

# Introduction

In this project, I performed topic modeling on scholarly abstracts available at Lens.org in order to get a better grasp on the type of scholarly research being conducted on generative AI. As this represents a technology that is changing rapidly and being actively incorporated into business strategies, the application of text analysis on the selected corpus may give some insight into where such developments are chiefly concentrated.

## Data

- Obtained from Lens.org
- 1,000 scholarly abstracts
- Query: “generative AI” OR “genAI”

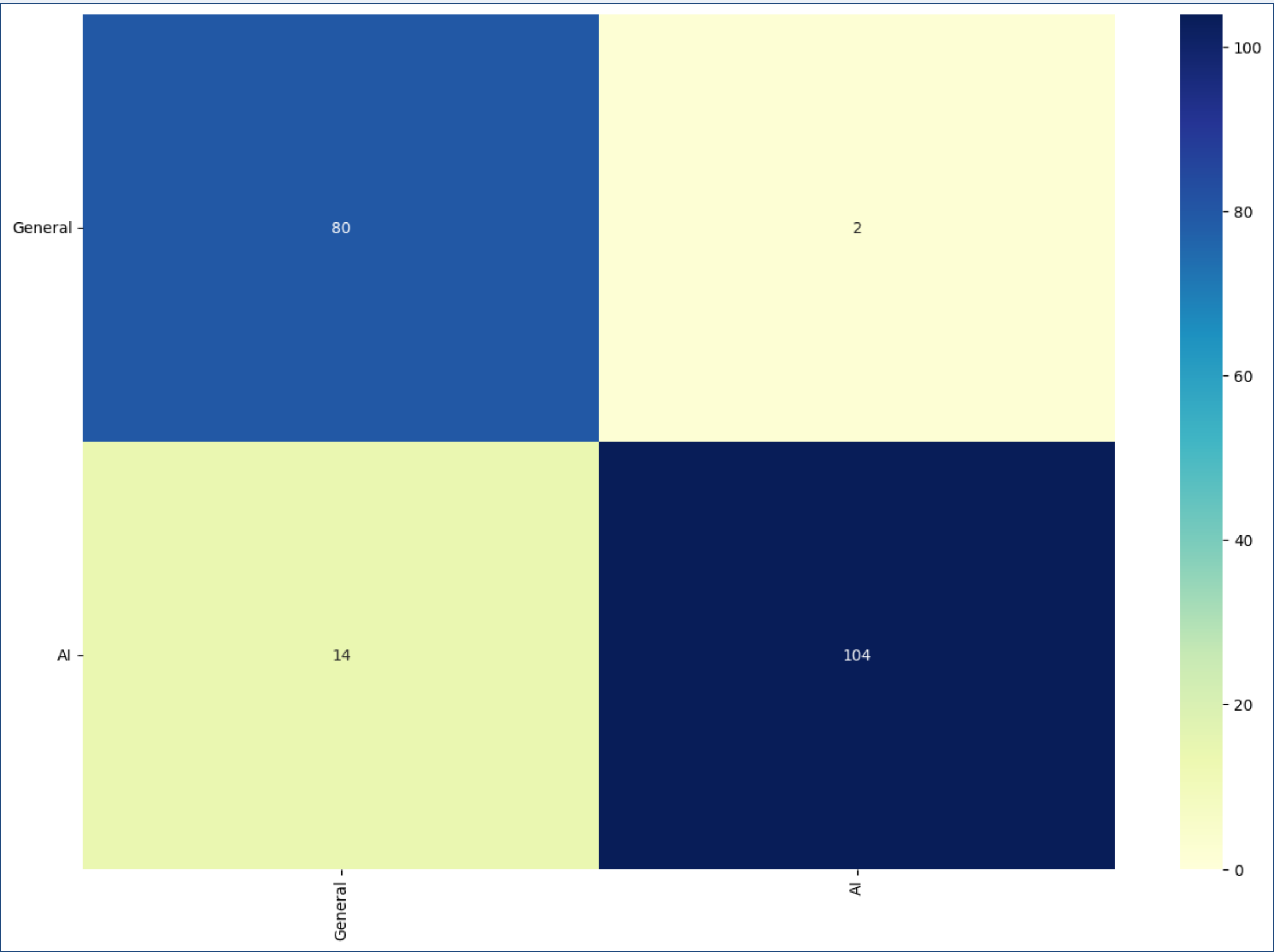
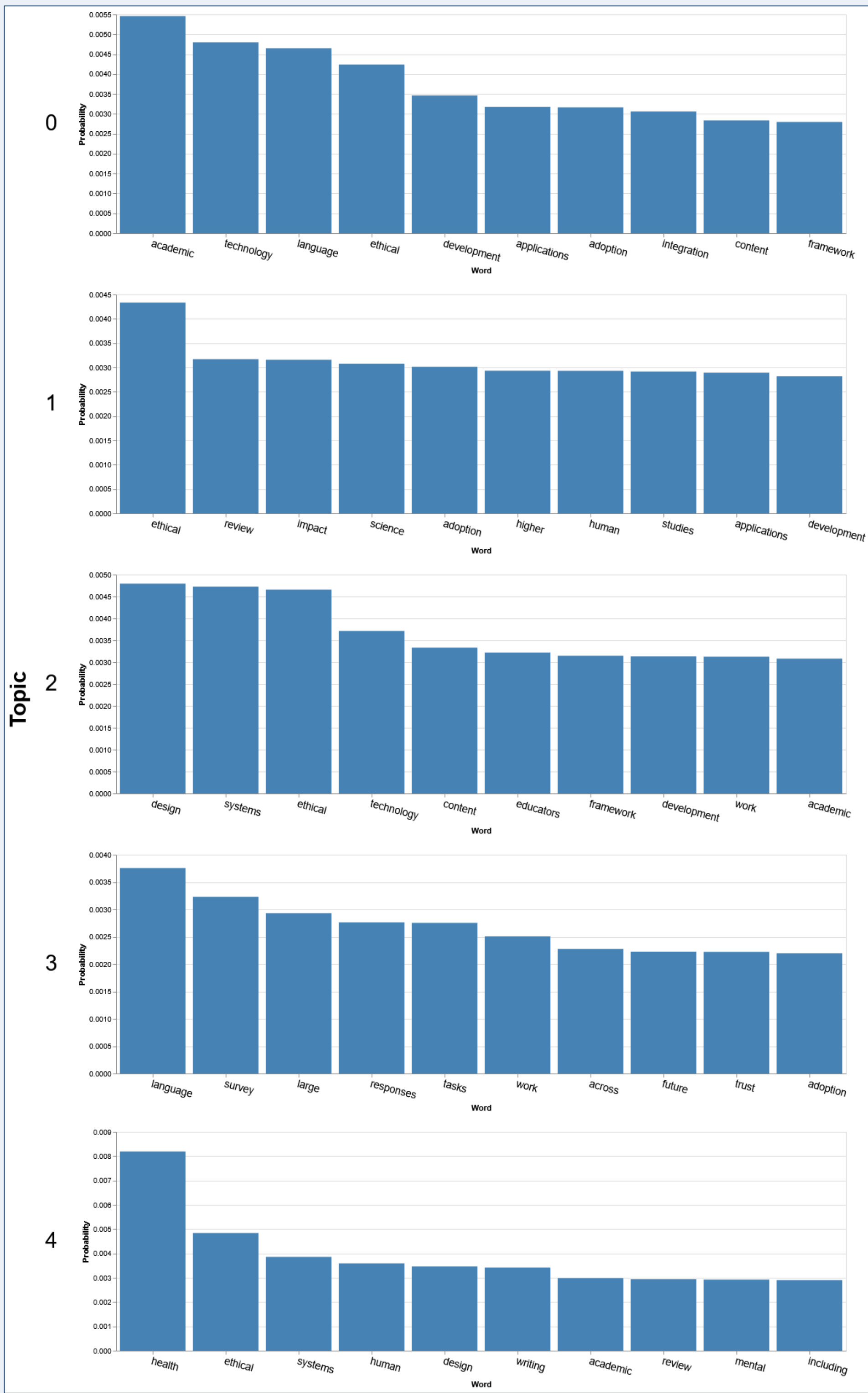
## Methodology

- Preprocessing removes punctuation, converts to lowercase, and splits abstracts into individual words.
- Latent Dirichlet allocation (LDA) is used to determine common themes present within the abstracts.
- Altair visualization is able to make the probability distribution of top words more apparent.
- Abstracts are designated as features, while labels are assigned to the dataset manually to perform classification based on whether an article covers a special topic or is more general.
- Different classifiers such as Naive Bayes, Logistic Regression, Support Vector Machine, AdaBoost, Random Forest are evaluated to build the best machine learning model.

## Text Analysis of Scholarly Abstracts at Lens.org: Generative AI

LIS 4693: Information Retrieval and Text Mining

Kane Peters; University of Oklahoma



# Findings

- Topic 0: GenAI in the academic environment
- Topic 1: Ethical issues and genAI
- Topic 2: Design of systems using genAI
- Topic 3: Large language models
- Topic 4: GenAI and health

Overlapping is present in the topics due to the specialized corpus.

The dataset was split using an 80/20 train-test split. Out of the classifiers used, AdaBoost achieved 92% accuracy and had the most favorable evaluation metrics overall.

## Conclusion

The project was able to gain some insight into the nature of scholarly research being conducted in the domain of generative AI. It uncovered various probable themes concerning the 1,000 document sample, such as education, healthcare applications, and ethical considerations.

In building a machine learning model, it may be possible to classify further corpora of similar documents in order to determine whether or not their scope leans toward dealing with generative AI in a specific context or a more generalized context.

## References

GitHub repository:

<https://github.com/kanepeters/lis4693/tree/main/final-project>

Dataset:

<https://github.com/kanepeters/lis4693/blob/main/final-project/lens-export%20FP.csv>